

Kaggle - Cafe Sales

Jacqueline Molina | June 2026

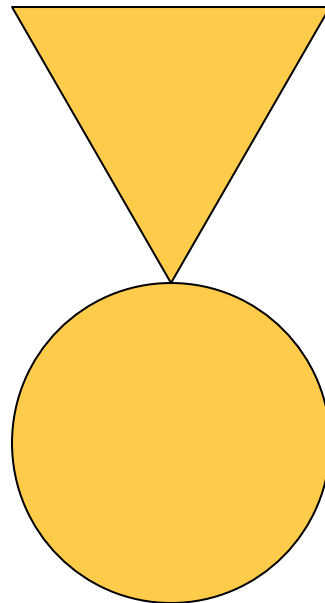


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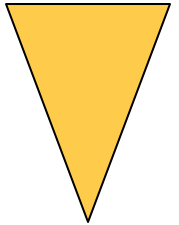
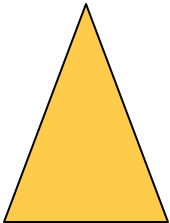
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Executive Summary

Analysis reveals that **Coffee, Juice, Cake** are the top three performers in quantities sold, but not in revenue generated. [(3766, \$7532),(3681, \$11043),(3655, \$10965)]. **Cookie** and **Tea** are the worst performing items [(3427, \$3427),(3441, \$5161)]. Data shows that **June** was the Cafe's best performing month generating revenue up to \$7353.0 and that generated item revenue is highly dependent on **Month**.

Key Findings

- **June Insights:** top performing month breakdown
- **Quantity Sold vs Revenue Generated:** grouped by item & grouped by month

Recommended Actions

- **Routine Maintenance:** ensure digital systems are consistently working
- **Promotions & Marketing:** encourage customers to buy/spend more
- **Reduce Menu Items:** remove worst performing items or create seasonal menus



Objectives

Business Task

Extract insights and highlight emerging patterns in Cafe data. Find trends based on items sold including: quantity of items sold and revenue generated from each item. Delve into revenue generated per day/week/month.

Questions:

- Which items sold the best/worst?
- How much revenue did they generate?
- Best performing month? Insights?
- How much revenue was generated per month? In Total?

Data Sources

<https://www.kaggle.com/datasets/ahmedmohamed2003/cafe-sales-dirty-data-for-cleaning-training/data>

Source: Public data from Kaggle

File Name: dirty_cafe_sales.csv

Rows: 10,000

Columns: 8

Transaction ID, *Item* (name of item), *Quantity* (quantity of the item purchased), *Price Per Unit* (price of a single unit of the item), *Total Spent* (total amount spent on transaction), *Payment Method*, *Location*, *Transaction Date*

Cleaning

Initial shape: (10000, 8)

- Quantity, Price Per Unit, Total Spent, Item are Interconnected
- Replace any '*Unknown*' or '*Error*' values with NaN values
- Fill in missing data with variations of this equation:
$$df['Quantity'] = df['Quantity'].fillna(df['Total Spent'] / df['Price Per Unit'])$$
- Drop rows where values cannot be identified or calculated

- Remove duplicate rows (none in this case)
- Remove Transaction ID, as it will always be unique and is not needed here
- Generate three new columns for Day, Month, Year, generated from Transaction Date

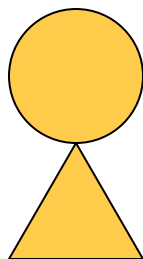
Final shape: (9514, 10)

Manipulation

- Total quantity of Items sold and total revenue generated per item
- Breakdown of revenue generated each month and quantity of items sold each month
- Created a subset of all transactions in the highest performing month (June)
 - Revenue generated daily
 - Highest performing day
 - Quantities of items sold on highest performing day (June 16)
- Average customer expenditure per month
- Box plot of total revenue generated per month (finding quartiles and ranges)
- Most/Least revenue generated in a day, Average Revenue generated, Average Customer Expenditure per day
- Most popular days of the week
- Sales of Items affected by Month (Line plot)
- Customer preferences of Location and Payment Method



Results & Recommendations



Data Studio: <https://datastudio.google.com/s/nwwOKvqqcec>

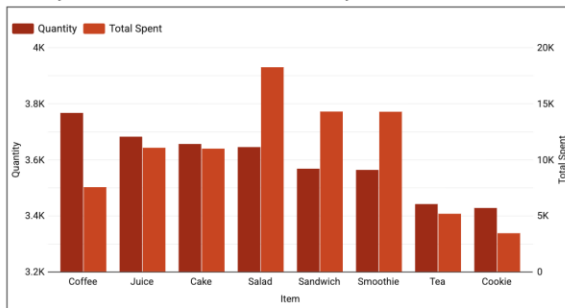
Tableau:

https://public.tableau.com/views/Cafe_VizTableu/JuneStats?:language=en-US&publish=yes&sid=&:redirect=auth&:display_count=n&:origin=viz_share_link

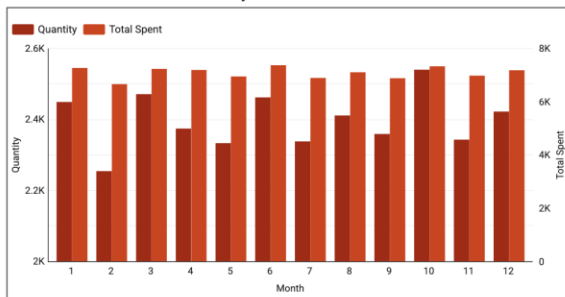
Key Findings



Quantity of Item Sold vs Total Revenue Generated by Item



Items Sold vs Revenue Generated by Month



Findings

- **Coffee, Cake, Juice** are top three performers in terms of quantities sold
- **Salad, Sandwich, Smoothie** are top three performers in terms of revenue generated
- **June, October, January** generated the most revenue
- **October, March, June** sold the most quantities of items

Implications

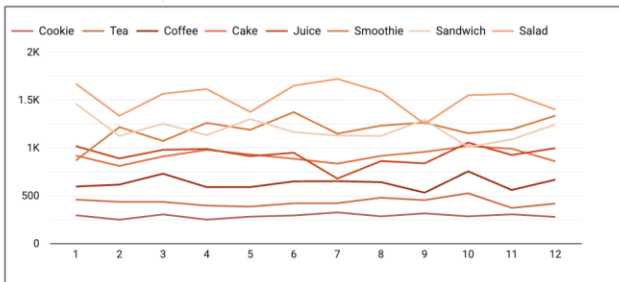
- Coffee, Cake, and Juice should be generating more revenue than they currently are seeing as they sell about the same as Salad, Sandwich, and Smoothie
- Revenue generated per month and quantities sold have no defined pattern or trend



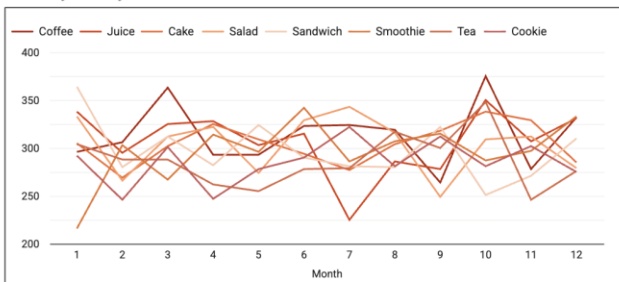
Key Findings



Revenue Generated by Month



Quantity Sold by Month



Findings

- **Cookie, Tea, Coffee** consistently generated the least amount of revenue throughout the year
- All items consistently generated about the same amount of revenue each month despite selling in different quantities
- Each item has its popular months and its unpopular months such as **Coffee in March** and **October**

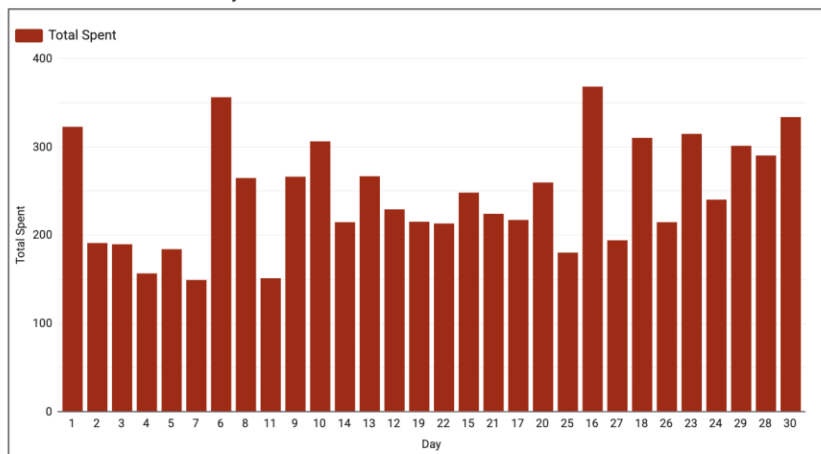
Implications

- Items which sold more quantities per month should be generating more revenue (not remaining about the same)
- Dynamically pricing menu items dependent on month would increase revenue



June Insights

Revenue Generated Each Day in June



Findings

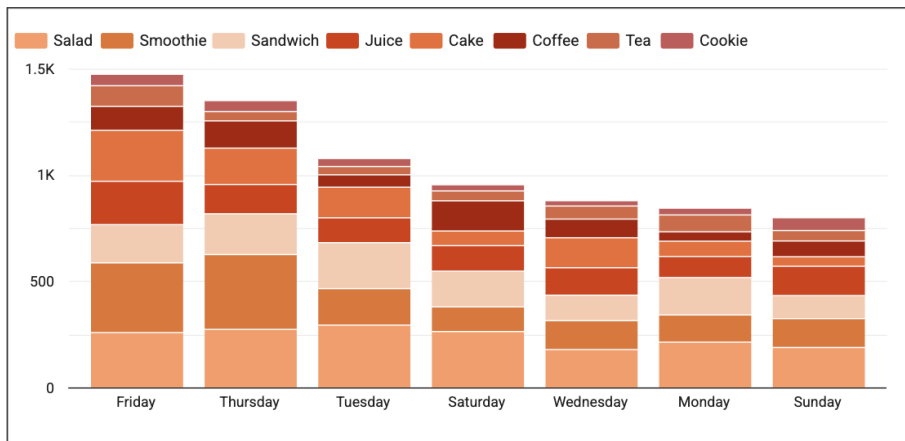
- Best performing day was June 16, 2023
- Worst performing day was June 7, 2023

Implications

- June had days where it generated over \$300 and it also had days where it generated less than \$200.

June Insights

Revenue Generated in June by Weekday



Findings

- **Salad** was always the most consistent in generating the most revenue, **Cookie** was always the worst
- **Friday, Thursday, and Tuesday** were the most popular days of the week in June

Implications

- Most revenue was generated on Friday's and Sunday's generated the least in June.

June Insights

HeatMap of Items Sold in June

Item / Quantity						
Top 30 - Day	Smoothie	Salad	Coffee	Juice	Cake	Grand total
1	32	9	11	12	1	102
2	5	1	13	19	10	85
3	-	5	22	14	9	69
4	10	7	8	5	6	55
5	13	11	10	3	-	62
6	15	25	2	9	14	101
7	15	-	8	5	8	51
8	19	10	14	10	10	86
9	10	5	4	14	32	90
10	6	26	13	5	9	87
11	1	3	10	26	3	61
12	10	6	1	6	7	86
13	8	7	17	11	4	85
Grand total	342	329	323	315	294	2,461

Findings

- **Smoothies** sold the best in June
- **Tea** sold the worst
- Popular days include **1, 6, 16, 18, 23, 28, 29, 30**

Implications

- Number of items sold were relatively the same with percentages ranging from 11.3% - 13.9%
- Each day had different popular items, no pattern emerged

Summary of Data Analysis

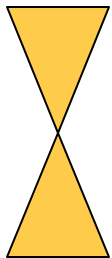


Findings

- **Top 3 Quantities Sold:** Coffee, Cake, Juice
- **Top 3 Revenue Generated:** Salad, Sandwich, Smoothie
- **June, October, January** generated the most revenue
- **October, March, June** sold the most quantities of items
- All items consistently generated about the same amount of revenue each month despite selling in different quantities
- Popular months for Items (**Coffee in March and October**)
- **Best** performing (revenue) day was June 16, 2023
- **Worst** performing (revenue) day was June 7, 2023
- **Salad** was always the most consistent in generating the most revenue, **Cookie** was always the worst
- **Friday, Thursday, and Tuesday** were the most popular days of the week in June
- **Smoothies** sold the best in June
- **Tea** sold the worst
- Popular days include **1, 6, 16, 18, 23, 28, 29, 30**



Summary of Data Analysis



Implications

- Coffee, Cake, and Juice should be generating more revenue than they currently are seeing as they sell about the same as Salad, Sandwich, and Smoothie
- Items that sold more per month should be generating more revenue (not remaining the same)
- June had days where it generated over \$300 and it also had days where it generated less than \$200.
- Most revenue was generated on Friday's and Sunday's generated the least in June.
- Number of items sold were relatively the same with percentages ranging from 11.3% - 13.9%
- Each day had different popular items, no patterns identified in June.



Recommendations

Reduce Menu Items

Remove the worst performing items from the menu
(Cookie/Tea) to prevent loss of profit or create a seasonal menu that introduces items based on past data month popularity

Promotions & Marketing

Introduce **dynamic** pricing.
Increase item prices in months where they are popular and decrease in less popular months.

Introduce promotions to encourage customers to buy less popular items by offering a 2-for-1 deal bundled with a popular item (e.g. **Coffee & Cookie**)

Provide discounts or promotions on popular days of the week such as **Thursdays**

Routine Maintenance

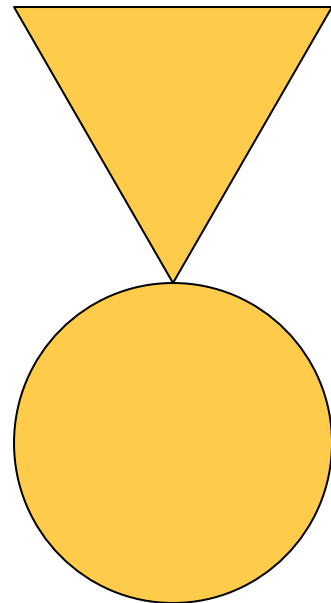
Assure all systems are in working order. The **majority** of customers prefer to pay through digital means. Having no access to technology results in loss of revenue.

Questions?

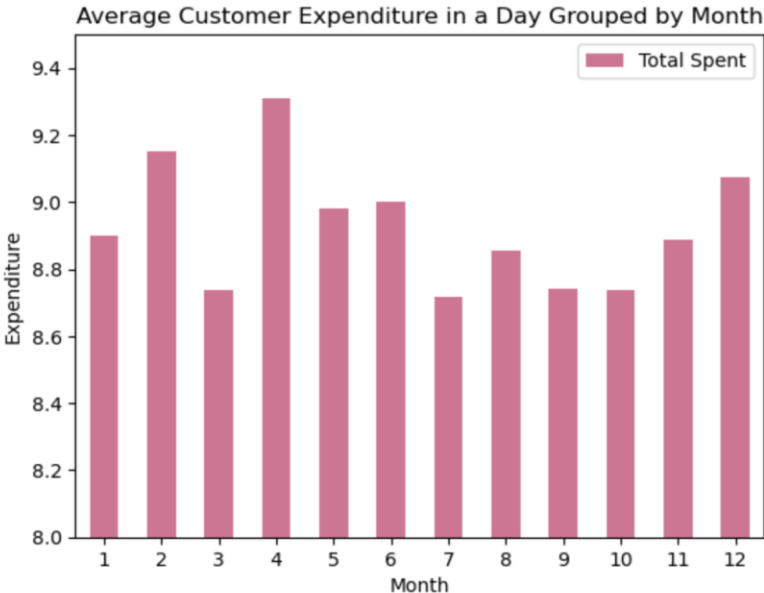
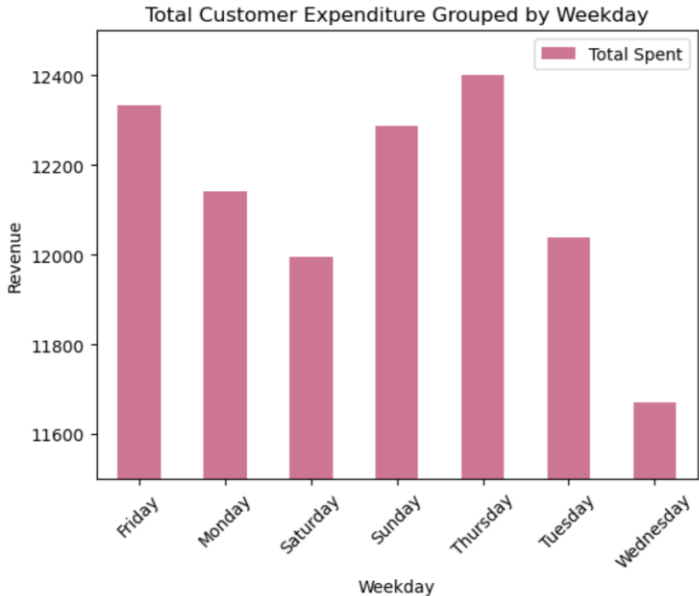
Github: <https://github.com/strwbrry05>

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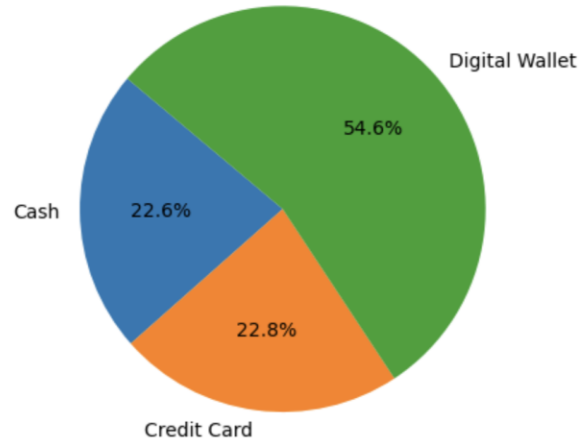


Appendix



Appendix

Ratio of Customers Preferred Payment Method



Ratio of Customers Preferred Dining Location

